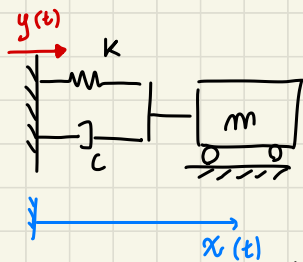




ESE 5

Useful shortcut when you have motion applied on the constrain of the system (forced constrain displacement)

(es)



using 1 DOF (x)

with excitation coming from a motion applied on the constrain

eq of motion

$$m\ddot{x} + c\dot{x} + Kx = c\dot{y} + Ky$$

this is equivalent to:

a force applied to the mass

$$f = c\dot{y} + Ky$$

proportional to speed and displacement of the clamp moving

known functions of time proportional to $y(t)$ respect K and $\dot{y}(t)$ respect c

GENERAL

equivalent to apply a moving force to the end of spring/damper system connected..

↳ constrain motion applied at one side of a spring/damper system

force defined by $[c\dot{y} + Ky]$ → related to spring stiffness and change of length y

faster solution than a 2 DOF system solution... using $\underline{x} = (\underline{x}_f \ \underline{x}_c)^T \dots$ require more time!

BUT this simplification DON'T WORK when:



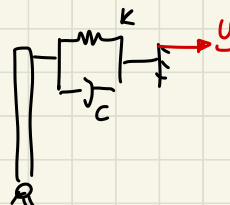
motion applied on the HINGE.

cannot be replaced by any linear force!

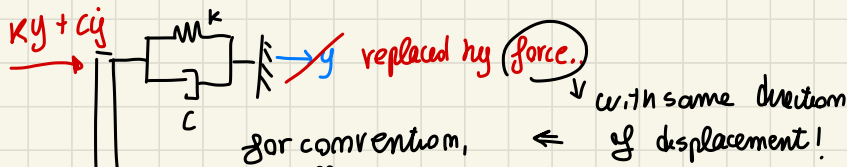
The actual respective excitation to y does NOT have only deformations BUT also moments...

NO shortcut

⇒ IF instead CONstrain MOTION applied directly to spring/damp syst



✓ YES, here you can apply shortcut ⇒



for convention,
usually $y(t) \geq 0$

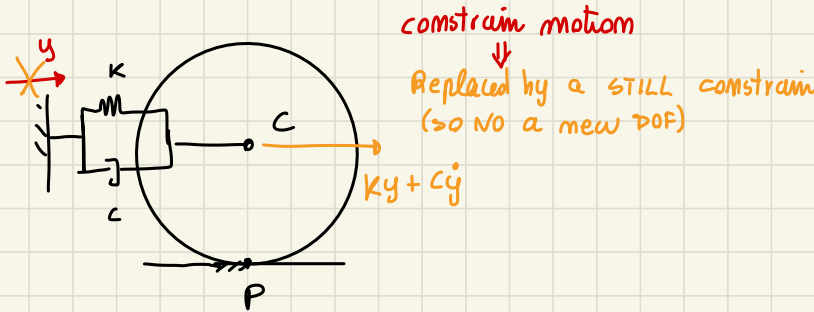
harmonic excitation, so $y, \dot{y}$ is a movement
changing direction

→ varying movement
of the constrain

" → " we are just
defining our positive
reference

typically we have harmonic
functions of time (or superimposition of it)

another example:



(NOTICE): on this shortcut, if we use a displacement

$$\begin{cases} y(t) = y_0 \cos(\Omega t) = \text{Re}\{y_0 e^{i\Omega t}\} \\ \dot{y}(t) = i\Omega y_0 e^{i\Omega t} \end{cases}$$

(EXP. function)
using complex
definition of $y(t)$
(easier to derive to
solve the diff eq.)

so

$$Ky + c\dot{y} = (\underbrace{K}_{\text{Re proportional to } K} + i\Omega \underbrace{c}_{\text{Im proportional to } \Omega, c}) y_0 e^{i\Omega t}$$

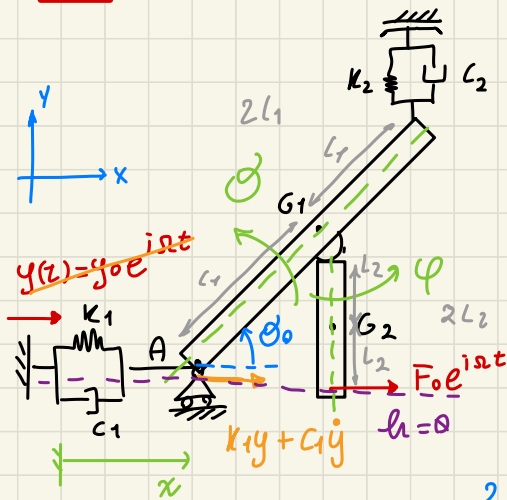
complex value
expression of equivalent
constrain force

⇒ NOT in phase with $y(t)$ movement,
phase antipathism due to the rate of $(\dot{y})$
deformation (90° in advance respect $y(t)$!)

so when dealing with constrain displacement, take the general solution
require more time than this easy shortcut!

ES P1 exam

2 Bar Rigid Body With uniform mass distribution



equilibrium position

2 sources of excitation } force
here we are in the } displacement on clamp

easy case when it is good
to take shortcut and

use a force

applied in opposite extremity
of spring/damping syst

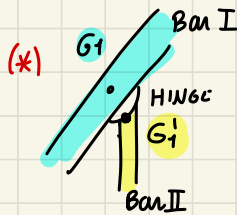
Solve the problem: by fundy DOF

2 R.B $\rightarrow$ 6 DOF

Cart A $\rightarrow$ 1 DOC

Hinge $G_1 \rightarrow$ 2 DOC $\Rightarrow$ connect two
(link two bodies) Bodies! (*)

3 DOF



$G_1 \in I$
 $G_1 \in II$

$x_{G_1} = x_{G_1'}$
 $y_{G_1} = y_{G_1'}$

2 D.O.C

2 kinematic condition
fixed!

$\neq$ different from
the case when

hinge in only one body
 $x_0 = 0$
 $y_0 = 0$ } still
2 D.O.C but
for different
reason

presume
one point
NOT moving

IF instead

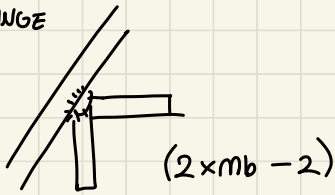
HINGE

2 bodies

$x_0 = 0$ $x_0' = 0$
 $y_0 = 0$ $y_0' = 0$

$(2 \times m_b) \overline{D.O.C}$

generalized by m bodies
connected to same HINGE



$(2 \times m_b - 2)$

IN total we shall consider 3 DOF small system to be described the instantaneous configuration by 3 independent kin. parameters
 of all system $\underline{x}$: horiz. position of point A

after locate A, we rotate BAR 1 to identify all positions bar

• $\odot$: we prefer to have indep coord

to be $\odot$ @ equilibrium! so our choice of $\odot$... it is BETTER (advantages) $\rightarrow$ respect canceling terms on average...

so our $\odot$ is measured respect static value $\odot_0$

so $\odot = \odot_0 + \odot$ when moving so @ equilibrium $\odot = 0$

• to find pendulum position we need ROTATION of BAR 2

$\odot$ measured from vertical position (STATIC equilibrium)

$\Downarrow$ $\underline{x} = \begin{Bmatrix} x \\ \odot \\ \varphi \end{Bmatrix}$ you can check that IF $x, \odot, \varphi$ FIXED, all system is STILL!
 $\hookrightarrow$ you can WRITE the equations following usual MATRIX LAGRANGE approach!

expressing properly T, V, D, W_{nc}

$$T = \frac{1}{2} \dot{\underline{y}}_m^T [m_{ph}] \dot{\underline{y}}_m \quad \text{We know } \dot{\underline{y}}_m = \begin{Bmatrix} V_{G1x} \\ V_{G1y} \\ \omega_1 \\ V_{G2x} \\ V_{G2y} \\ \omega_2 \end{Bmatrix} \quad [m_{ph}] = \text{diag} \left\{ m_1; m_1; J_1; m_2; m_2; J_2 \right\}$$

BUT usually we don't need all V, ω of Bodies...
 you can simplify expression
 HERE all components are relevant!

$\leftarrow$ has the center of mass linear velocity + ROTATION respect relevant velocity for T

$\uparrow$ KNOWN DATA

$\downarrow$ then we need to WRITE

$$\dot{\underline{y}}_m = \begin{bmatrix} \lambda m \end{bmatrix} \dot{\underline{x}}_{6 \times 3}$$

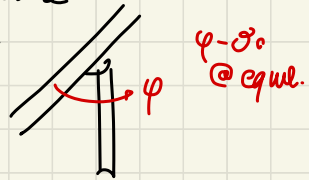
$\hookrightarrow$

$$[\lambda_{lm}] = \begin{bmatrix} \ddot{x} & \ddot{\theta} & \ddot{\varphi} \\ 1 & -L_1 \sin(\theta_0) & 0 \\ 0 & L_1 \cos(\theta_0) & 0 \\ 0 & 1 & 1 \\ 1 & -L_1 \sin(\theta_0) & L_2 \\ 0 & L_1 \cos(\theta_0) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} V_{G1x} \\ V_{G1y} \\ \omega_1 \\ V_{G2x} \\ V_{G2y} \\ \omega_2 \end{matrix}$$

NOTICE relative to $\dot{\varphi}$, you may consider that when dealing with connected bodies you should clarify IF φ is absolute rotation or Relative. φ ABS. if measured respect fixed y axis OR it could be chosen φ sometimes as REL. respect BAR 1, so it would be

you should approach the problem consistently with your φ choice of measurement $\rightarrow$ be sure if φ ABS or REL to be consistent in every measurement of φ !
HERE is easy / safe to see φ is ABSOLUTE!

{ Different? choices }



compute it super-imposing:

• $\ddot{x} \neq 0, \ddot{\theta} = 0, \ddot{\varphi} = 0$

IF not ROTATING body $\ddot{\theta} = 0, \ddot{\varphi} = 0$, so all points has same Vel, TRANSLATION, it is enough one point movement $\Rightarrow$ POINT (A) is enough!

$G_2 \equiv G_1$ moves, in same way!

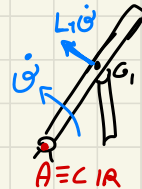
• $\ddot{x} = 0, \ddot{\theta} \neq 0, \ddot{\varphi} = 0$

JUST $\ddot{\theta}$ VAR4... Bar 2 translate $\ddot{x}$ fixed motion, it is like replay CART with HINGE

early $A \equiv CIR$

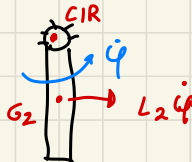
We have horiz/vert comp of V_{G1}

φ frozen, so G_2 moves as G_1



• $\ddot{x} = 0, \ddot{\theta} = 0, \ddot{\varphi} \neq 0$

IT is obviously that BAR 1 is frozen
While Bar 2 (pendulum) moves as
(purely horizontal motion)



from $[m_{ph}], [\lambda_m] \rightarrow [m] = [\lambda_m]^T [m_{ph}] [\lambda_m]$ total mass matrix

V) Contribution of el + grav. sometimes $V_g, V_{el II}$ due to pre-load only in K . We have NOT HERE cancel!

$\Delta e_1 = x$ linear spring NOT (K_{II}) BUT K_2 has non lin. relation with θ + NOW IN Relationship on G_1, G_2 terms related to respect $\theta, \varphi \dots [K_g] \neq 0$

cons. forces is difficult than usually! $\rightarrow$ we shall consider both $[K_I] + [K_{II}] + [K_g]$ — NONZERO grav stiff BOT for B1/B2!
second order stiff due to K_2 (while K_1 linear) only SPRING 2)

$V_{el} =$ FIRST ORDER..

$$= \frac{1}{2} \underline{y}_k^T [K_{ph}] \underline{y}_k$$

$$\underline{y}_k = \begin{Bmatrix} \Delta e_1 \\ \Delta e_2 \end{Bmatrix} [K_{ph}] = \text{diag} \{K_1, K_2\} = \begin{bmatrix} K_1 & 0 \\ 0 & K_2 \end{bmatrix}$$

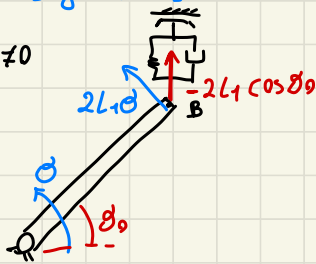
$$\downarrow \underline{y}_k \approx [\lambda_k] x$$

approximate @ I ORDER elastic terms

$$[\lambda_k] = \begin{bmatrix} x & \theta & \varphi \\ +1 & 0 & 0 \\ 0 & -2L_1 \cos \theta_0 & 0 \end{bmatrix} \begin{matrix} \Delta e_1 \\ \Delta e_2 \end{matrix}$$

$\leftarrow + \rightarrow$
(convention) { filled with super imposition approach as for (λ_m) }

When $\theta \neq 0$



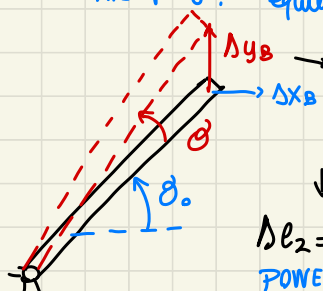
If $\dot{\varphi} \neq 0$ all BARI frozen, NO changes on springs!

to consider the II ORD stiffness for SPRING 2!

Find a non lin expression for Δl_2 to decompose in POWER SERIES expansion ... highlight II ORD term!

$$\Delta l_2 = -\Delta y_B + \Delta l_{02}$$

↑
if $y_B > 0$
it compress
the spring!
STATIC
displacement,
depending on
equilibrium posit. of $\Delta y_B(0)$ (PRE-LOAD)



$$\Delta y_B = 2L_1 \sin(\theta_0 + \theta) - 2L_1 \sin(\theta_0) = 2L_1 (\sin(\theta_0 + \theta) - \sin(\theta_0))$$

$$\Delta l_2 = -2L_1 (\sin(\theta_0 + \theta) - \sin(\theta_0)) + \Delta l_{02}$$

POWER SERIES expansion of it...

JUST depending on θ (Lucky)

We can express it by cos, sin POWER exp mean $\theta = 0$
(NOTABLE RESULT)

should rewrite it...

$$\Delta l_2 = -2L_1 (\cos(\theta_0) \sin(\theta) + \sin(\theta_0) \cos(\theta) - \sin(\theta_0)) + \Delta l_{02}$$

$\approx$ NEAR $\theta = 0$
 θ

$\approx$ NEAR $\theta = 0$
 $1 - \frac{\theta^2}{2}$ McLaurin Expansion

$$= -2L_1 (\theta \cos(\theta_0) + \sin(\theta_0) - \frac{\theta^2}{2} \sin(\theta_0) - \sin(\theta_0)) + \Delta l_{02} =$$

$$= \Delta l_{02} - (2L_1 \cos(\theta_0)) \theta + (2L_1 \sin(\theta_0)) \frac{\theta^2}{2}$$

[Kel I] for SPRING 2
respect θ , as FOUND KED

DATA of
PROBLEM

$$K_{el II, 2} = K_2 \Delta l_{02}$$

↑
only respect
SPRING 2

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2L_1 \sin(\theta_0) & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

TOTAL
HESSIAN
MATRIX

Δl_2 depends only
on θ , all other
terms respect φ, x are 0

GRAVITATIONAL POTENTIAL energy (GRAVIT. stiffness matrix)

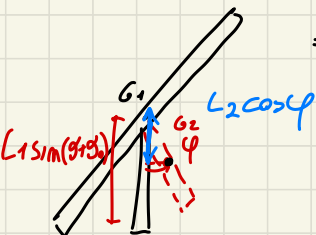
$$V_g = m_1 g \underbrace{h_1}_{\text{height of } G_1} + m_2 g \underbrace{h_2}_{G_2 \text{ height}} =$$

Use as REF level
the horiz axis passing
through A

$$= m_1 g L_1 \sin(\theta_0 + \theta) + m_2 g (L_1 \sin(\theta_0 + \theta) - L_2 \cos(\varphi)) =$$

$$= m_1 g L_1 (\cos(\theta_0) \sin(\theta) + \sin(\theta_0) \cos(\theta)) + m_2 g (L_1 \cos(\theta_0) \sin(\theta) + L_1 \sin(\theta_0) \cos(\theta) - L_2 \cos(\varphi))$$

LINEAR I $\approx \theta$
II ORDER APPROX ≈ 0
COSTANT TERMS $\approx 1/\theta^2/2$
 $\approx 1/\varphi^2/2$



WHAT
REMAIN..

We care only about II ORD -- because when derived, constant goes to 0, while I ORD will cancel with elastic constant
I ORD even if mom & cancel with first ORD terms of elastic potential energy precisely
K1, K2 (DON'T CARE ABOUT)

through away
with I ORD elastic STIFFNESS, to cancel on total Lagrange equation

$$[K_g] = m_1 g \begin{bmatrix} 0 & 0 & 0 \\ 0 & -L_1 \sin(\theta_0) & 0 \\ 0 & 0 & 0 \end{bmatrix} + m_2 g \begin{bmatrix} 0 & 0 & 0 \\ 0 & -L_1 \sin(\theta_0) & 0 \\ 0 & 0 & L_2 \end{bmatrix}$$

negative contribution

because create instability when $[K_{tot}]$ has NEG term

because $C_1 // K_1$, $C_2 // K_2$ the DUMPING MATRIX is some of $[\lambda_K]$
linking $\dot{x}$ to $\dot{y}$
↓

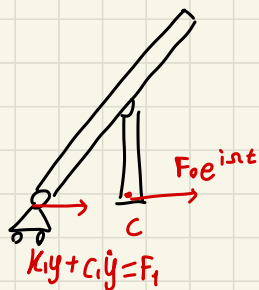
So we will use some JACOBIANS of λ_K
by $[C_{ph}]$ matrix, C_1, C_2 instead K_1, K_2

ending by CONSIDERING VIRTUAL WORK..

we have 2 time dep. forces applied

$$\delta W_{nc} = \delta \underline{y}_F^T \underline{f}_{ph} \quad \underline{f}_{ph} = \begin{Bmatrix} F_0 e^{i\omega t} \\ F_1 \end{Bmatrix}$$

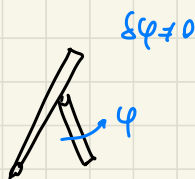
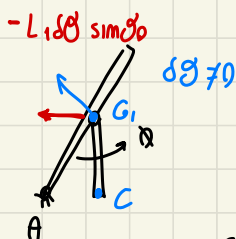
$$\delta \underline{y}_F = \begin{Bmatrix} \delta x_c \\ \delta x_A \end{Bmatrix} = \downarrow = [\lambda_F] \delta \underline{x}$$



superimposes
the cases

$$[\lambda_F] = \begin{bmatrix} \delta x & \delta \theta & \delta \varphi \\ 1 & -L_1 \sin \theta_0 & 2L_2 \\ -1 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \delta x_c \\ \delta \theta \\ \delta \varphi \end{Bmatrix}$$

↑
cell moving
w/ δx
TRANSLATING



SO we have all the TERMS to obtain the explicit
(LAGRANGE ER by $[m], [c], [k], \underline{f}$)

TWO EFFECTS that
can be considered using

reflects the horizontal force
applied to C + constraint
motion

super-imposition, principle (singularly apply forces)

Different actg inputs analysis!